



FRUIT BAGGING: SHIELD TO YIELD WITH QUALITY FRUITS

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Abstract

Fruits play a major role in the daily diet of the human and are major sources of various vitamins and minerals as well as they provide a very good income to the farmers. Several environmental factors have a significant role during fruit growth and their development. Many pre-harvests biotic (diseases, pests, birds) and abiotic (include genetic, environmental and cultural) stresses influence the postharvest quality of the fruits and lead to susceptibility of fruits to diseases and various physiological disorders like cracking and sunburn.

Keywords: Fruit bagging, Types of bags and Techniques

Introduction:

Bagging refers to the covering of fruits with bags to protect them from various biotic and abiotic factors. It is a technique, which provides physical protection to the fruits, which helps in improving their physical and internal quality as well as changes the microclimate inside the bag for proper growth and development



History: The bagging technique, which was first utilized in Japan in the 20th century for pears and grapes, is now widely applied in Asian countries (Japan, China, Korea), Australia and the USA.

Importance of fruit bagging

- 1.Enhance fruit quality:** Fruit bagging enhances internal quality by improving the sugar content, flavour and nutritional value of the fruit.
- 2.Improves physical appearance:** It also improves physical appearance by preventing blemishes, sunburn and mechanical damage.
- 3.Pest and diseases:** Its effective protection from pest and diseases by acting as physical barrier against insects and birds.
- 4.Environmental stress:** Also minimizes environmental stress by shielding fruits from harsh weather climatic conditions like excessive sunlight, wind and rain
- 5. Market value:** It significantly increases the market value
- 6.** It promotes eco-friendly farming practice.

Important steps involved in fruit bagging

1. Timing - After fruit set and initial growth.
2. Methods - Completely enclosed without causing damage.
3. Securing - To prevent them from falling off or being dislodged by wind or rain.
4. Bag size- Appropriate bag size ensures proper fruit development without affecting the growth.

Bagging material

Breathability: Ensures air flow for fruit health

Strength and durability: By withstand environment stress

UV protection and Biodegradability: Safeguarding fruit and promoting eco-friendly

Different types of bagging:

- 1)Dried banana leaves
- 2)Jute bag
- 3)Newspaper bag (NPB)
- 4)White paper bag (WPB)
- 5)Yellow paper bag (YPB)
- 6)Transparent polythene
- 7)Blue polythene
- 8)Red polythene
- 9)Black polythene
- 10)Cellophane paper

Monitoring: Regular inspection is done to check the conditions of the bags and fruits then any damaged bags are placed to protect the fruits and finally the fruits are harvest when they are fully ready for harvest.

Key steps

Management before bagging - Involves preparing the fruit and tree by selecting healthy fruits, thinning excess fruits and applying necessary treatments like pest control or fungicides.

Bagging- This is the step where selected fruits are covered with protective bags to shield them from pests, diseases, sunburn and mechanical damage

Management after bagging - After bagging, proper care such as irrigation, nutrition, and monitoring for pests or diseases continues to ensure healthy fruit development inside the bags.

Removal of bags - Bags are carefully removed at the appropriate time before harvest to allow fruits to develop their natural colour and improve market appearance.

Management after bag removal - After removing bags, fruits are exposed to the environment, so continued protection and care are essential to maintain their quality and appearance.

Fruit harvest - Finally, the mature and high-quality fruits are harvested for consumption or market sale.

Benefits of fruit bagging

- It ensures uniform fruits ripening with attractive coloration and quality
- Certain bagging materials can be reused again in the next fruiting season
- Pre-harvest bagging protect the fruits from the pest and disease infections
- High recovery of marketable fruits
- It protects fruits from damage by bruises arising due to strong winds, birds attack and physical injuries

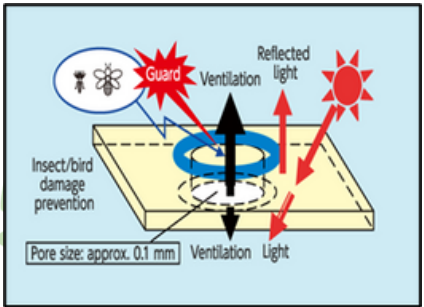
Special techniques

Bag with an umbrella: For fruit protection especially against sunburn over the fruit and the small umbrella on top gives extra shade it can be closed or opened depends on the sunlight difference fruits like grapes and pears being protected with these umbrella style bags which helps reduce direct sunlight improve fruit quality.



Microporous bags

Used in fruit protection which have a pore size of approximately 0.1mm these tiny pores allow air to pass through providing ventilation while blocking insects’ birds and parasite the bags also help in preventing damage and offers diffused light reflection reducing the chance of sunburn they are light making them easy to handle without affecting fruit growth.



Fruit bagging machine

This automatic fruit bagging machine packs up to 500 bags per hour with electric heat sealing, saving time and labor for farmers.



Conclusion

Fruit bagging is an effective and eco-friendly technique that acts as a protective shield, improving both the quantity and quality of fruits. By reducing the need for chemical pesticides and improving market value, bagging supports sustainable farming and ensures safe, high-quality produce for consumers. Thus, fruit bagging is a smart and simple tool to yield better returns with superior quality fruits.

References

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